PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effect of computerised, knowledge-based, clinical decision support systems on patient-reported and clinical outcomes of patients with chronic disease managed in primary care settings: a systematic review
AUTHORS	El Asmar, Marie line; Dharmayat, Kanika; Vallejo-Vaz, Antonio J.; Irwin, Ryan; Mastellos, Nikolaos

VERSION 1 – REVIEW

REVIEWER	Belmin, J Hôpital Charles Foix et Université Pierre et Marie Curie Paris, Geriatrics
REVIEW RETURNED	11-Jul-2021

GENERAL COMMENTS	This manuscript by El Asmar et al reports a systematic review which has investigated the effects of computerized decision support system on patients' outcome in primary care settings. The systematic review has been conducted on a limited number of chronic diseases. The topic is relevant and the methods are appropriate. Major points: Authors should point to previous systematic reviews that have examined the effects of CDSS on chronic disease, and use them to identify relevant studies and also to discuss the results of the present study. For example, in 2013, Cleveringa et al published a valuable systematic review on diabetes care and patient outcomes in Diabetes Technology and Therapeutics. They found 12 studies that had examined the effects of CSSD on diabetes control in the primary care setting. Why were these studies not included in this systematic review? There is an overlap between the content of Tables 1 and 3, particularly for "patient identification" and participants. The authors should resolve this and avoid redundancy. The term "patient identification" is not relevant and should be modified. In Table 1 and 4, the study by Eccels is described as targeting asthma and angina. In Table 2, the authors should explain how the CDSS try to improve management of these two diseases. In Table 4, for the study by Cobos, there is a problem by reporting the confidence intervals of the ratio that do not contain the OR. Also, results in the absolute effect columns are unclear. Are these baseline values or treatment effects? In the study by Hicks, could the authors might present the results as differences in outcome, as it is done with other studies. In Table 4, the treatment effects is not understandable easily. Particularly, when expressing differences, readers should be able to understand clearly if the intervention reduced or increased the HbA1c or reduced or increased blood pressure. Maybe a sentence

	in Methods could clarify how are calculated the differences. In Results, the section entitled narrative synthesis, authors should choose either to present each study in detail or to present a group of studies dealing with a specific disease, but not to alternate between the two types of presentation. This section contains a large number of interpretations, which are not appropriate in the Results. These points should be retained for Discussion or deleted. In addition, authors used to term “non-superiority” to describe the effects of interventions of two studies. This is not appropriate and authors should use more conventional ways to reports the results. In Results, the authors stated that they did not perform a meta-analysis because the reported outcomes were not suitable for pooling. However, in the diabetes studies, all had HbA1c and cholesterol among the outcomes, and in the hypertension studies, all had blood pressure measurements. It is more likely that the heterogeneity of the studies may justify not conducting a meta-analysis. In Discussion, the authors rose quite negative opinions about the effects of CDSS in the studies they have included. I found that the effects on hypertension control are quite interesting in the study by Lopez and that the increase in the proportion of hypertensive patients who achieve goals are clinically relevant. Minor points: In Table 4, the presentation of the confidence intervals should be changed to avoid confusion. The “- “ between the lower and upper limits should be replaced with “to”. For the study by Gill, avoid reporting goal HbA1c<9% because it is not listed in outcomes column. Avoid repetition of the definition of the goal in the column Relative effects, since it is detailed in column Outcomes. In the Tables, the reference number of the studies should be provided.

REVIEWER	Cheerkoot-Jalim, Sudha University of Mauritius, Information and Communication Technologies
REVIEW RETURNED	26-Jul-2021

GENERAL COMMENTS	The methodology used for the systematic review is commendable, with the application of the PRISMA guidelines, the risk of bias assessment and the quality assessment of outcomes. CDSS’s are widely adopted across many areas in the healthcare domain and it would be interesting to read about the findings of a systematic review assessing their effect on patient outcomes. However, the major concern here is the number of articles included in the final review, which is largely insufficient to report findings and deduce conclusions on the topic. Further analysis and deductions based on the sub-domains (like T2DM) becomes even more challenging. Moreover, given the rate at which CDSS’s are evolving, it would be expected to include more recent studies for such type of review. However, the current review includes very few recent studies. Corrections required: • In Abstract, it was mentioned that the studies were searched up to 24th June 2021. However in the Methods Section, it is mentioned that the databases were searched from inception to 24th June 2020 • P7 line 56: mistype ‘ovidence’ • There are two Supplementary Table 3

VERSION 1 – AUTHOR RESPONSE

Reviewer 1		
	Comments	Responses
1	Authors should point to previous systematic reviews that have examined the effects of CDSS on chronic disease, and use them to identify relevant studies and also to discuss the results of the present study. For example, in 2013, Cleveringa et al published a valuable systematic review on diabetes care and patient outcomes in Diabetes Technology and Therapeutics. They found 12 studies that had examined the effects of CSSD on diabetes control in the primary care setting. Why were these studies not included in this systematic review?	• All systematic reviews in the field were hand-searched for potential studies that could fit the PICOS criteria. • Regarding the study by Cleveringa et al., the included studies did not fit our PICOS criteria. These were web-based, not used by PCP (other healthcare professionals), not linked to EHR, some did not compare to usual care not utilizing CDSS, non-electronic reminders, multifaceted interventions, printed reminders.
2	*There is an overlap between the content of Tables 1 and 3, particularly for “patient identification” and participants. The authors should resolve this and avoid redundancy. The term “patient identification” is not relevant and should be modified.	Thank you for noting this. Tables 1 and 3 have been merged.
3	In Table 1 and 4, the study by Eccels is described as targeting asthma and angina. In Table 2, the authors should explain how the CDSS try to improve management of these two diseases.	Table 2 is amended and includes details on both angina and asthma.
4	*In Table 4, for the study by Cobos, there is a problem by reporting the confidence intervals of the ratio that do not contain the OR. Also, results in the absolute effect columns are unclear. Are these baseline values or treatment effects? In the study by Hicks, could the authors might present the results as differences in outcome, as it is done with other studies.	• The absolute effects column represents differences in lab values post-intervention compared to baseline (pre-intervention) for each of the intervention and usual care groups individually, whereas the relative effects column represents between- (intervention and usual care) group differences post-intervention. • Regarding the study by Hicks, as defined by the ‘outcomes’ column, the differences in Systolic and diastolic blood pressures (SBP and DBP) are already presented.
5	*In Table 4, the treatment effects is not understandable easily. Particularly, when expressing differences, readers should be able to understand clearly if the intervention reduced or increased the HbA1c or reduced or increased blood pressure. Maybe a sentence in Methods could clarify how are calculated	For ease of understanding the results and their interpretations, this has been directly clarified in the table for each reasult presented where this was not clear.

	the differences.	
6	In Results, the section entitled narrative synthesis, authors should choose either to present each study in detail or to present a group of studies dealing with a specific disease, but not to alternate between the two types of presentation. This section contains a large number of interpretations, which are not appropriate in the Results. These points should be retained for Discussion or deleted.In addition, authors used to term "non-superiority" to describe the effects of interventions of two studies. This is not appropriate and authors should use more conventional ways to reports the results.	• The narrative synthesis section is structured as follows: studies are first grouped by disease condition, and then, within each disease condition, each study is presented in detail. We thank the reviewer for highlighting this point. We have reviewed the results section to avoid interpretations where this clearly occurred and so presented the results more objectively. • The amended to remove interpretations, or perceptions of interpretation, and the section around 'non-superiority' has been removed given the reviewer's comments.
7	In Results, the authors stated that they did not perform a meta-analysis because the reported outcomes were not suitable for pooling. However, in the diabetes studies, all had HbA1c and cholesterol among the outcomes, and in the hypertension studies, all had blood pressure measurements. It is more likely that the heterogeneity of the studies may justify not conducting a meta-analysis.	We have amended the text to indicate the reason for not pooling the data in a meta-analysis as suggested.
8	*In Discussion, the authors rose quite negative opinions about the effects of CDSS in the studies they have included. I found that the effects on hypertension control are quite interesting in the study by Lopez and that the increase in the proportion of hypertensive patients who achieve goals are clinically relevant.	The text has been revised to adjust the wording.
1	In Table 4, the presentation of the confidence intervals should be changed to avoid confusion. The "- " between the lower and upper limits should be replaced with "to". For the study by Gill, avoid reporting goal HbA1c<9% because it is not listed in outcomes column.Avoid repetition of the definition of the goal in the column Relative effects, since it is detailed in column Outcomes.	Confidence intervals resolved/edited to 'to'. • Regarding the study by Gill, the •'outcomes column' is adjusted to reflect all the outcomes reported • The repeated text which occurs in 'definition of the goal' in the 'outcomes' column in the 'relative effects' column is to highlight to the user which effects are being described rather than repeating for the purposes of 'defining the goal'
2	In the Tables, the reference number of the studies should be provided.	Reference numbers of the studies is now included in the tables.

Reviewer 2		
	Comments	Responses
1	In Abstract, it was mentioned that the studies were searched up to 24th June 2021. However in the Methods Section, it is mentioned that the databases were searched from inception to 24th June 2020	The date has been amended
2	P7 line 56: mistype 'ovidence'	The mistake has been edited
3	There are two Supplementary Table 3	The mistake has been revised.

VERSION 2 – REVIEW

REVIEWER	Belmin, J Hôpital Charles Foix et Université Pierre et Marie Curie Paris, Geriatrics
REVIEW RETURNED	13-Nov-2021

GENERAL COMMENTS	This manuscript reports a systematic review of the effects of computerized clinical decision support systems on patient outcomes in primary care. The subject is of great interest and relevant for clinical practice. This systematic review is well conducted with appropriate methodology and the results are well presented. Table 2 is a strength of the manuscript, as in many articles CDSS are poorly described. No major critical points. Minor points: The legend to Table 3 should mention the effects of CDSS on patient health outcomes. Table 1 should be condensed to improve readability. For example, the country column should be removed, and countries should be added in the first column with the first author and year. The outcome column is redundant with the one in Table 3. In Table 1, the authors should replace it with an objective or target column (indicating a simplified outcome). The detailed outcome could be remain detailed in the Table 3.

VERSION 2 – AUTHOR RESPONSE

Reviewer's comments	
1) The legend to Table 3 should mention the effects of CDSS on patient health outcomes.	• Legend of table 3 has been adjusted as suggested
2) Table 1 should be condensed to improve readability. For example, the country column should be removed, and countries should be added in the first column with the first author and year. The outcome column is redundant with the one in Table 3. In Table 1, the authors should replace it with an objective or target column (indicating a simplified outcome). The detailed	• Table 1 has been condensed by deleting the 'country' column and listing the countries under the first author and year column. • The 'outcome' column has been 'target outcomes' and the listed outcomes are presented in a simplified format with detailed

outcome could be remain detailed in the Table 3.	outcomes presented in table 3